

WHY DO YOU LIKE FUNCTIONAL
PROGRAMMING SO MUCH? WHAT
DOES IT ACTUALLY *GET* YOU?

TAIL RECURSION IS
ITS OWN REWARD.



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@bitsapien / Rahul C





NOTHING



A dose of Elixir



What's in for you in this
talk?

What's in for you in this
talk?

HUNGRY

What's in for you in this talk?

HUNGRY

COOL FEATURES





pure joy

```
ScalaFiddle.scala:2: error: not found: value println  
println("Hello world!")  
^
```



pure joy

```
ScalaFiddle.scala:2: error: not found: value println  
  println("Hello world!")  
  ^
```



pure joy

```
undefined method `prit' for main:Object  
Did you mean?  print  
(repl):1:in `'
```



















Aha! 🧠



+



+





Why is Elixir awesome?

Why is Elixir awesome?

Programmer First

Why is Elixir awesome?

Programmer First

```
✖ Uncaught TypeError: Class extends value undefined is not a constructor or null game.js:4961  
  at Object.<anonymous> (game.js:4961)  
  at __webpack_require__ (game.js:4723)  
  at Object.<anonymous> (game.js:5388)  
  at __webpack_require__ (game.js:4723)  
  at Phaser.Group.progress (game.js:4769)  
  at Object.286 (game.js:4772)  
  at s (game.js:1)  
  at game.js:1  
  at Object.293.babel-runtime/core-js/object/get-prototype-of (game.js:6706)  
  at s (game.js:1)
```

```
iex (beam.smp)
iex(1)> clear
iex(2)> IO.put "1"
** (UndefinedFunctionError) function IO.put/1 is undefined or private. Did you mean one of:
    * puts/1
    * puts/2

(elixir) IO.put("1")
iex(2)> 
```

→ Sessions mix new partialBBQ --sup

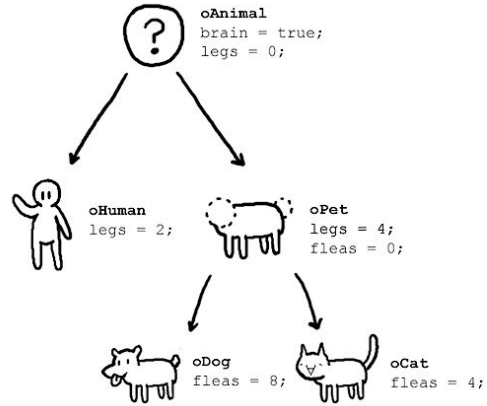
** (Mix) Application name must start with a lowercase ASCII letter, followed by lowercase ASCII letters, numbers, or underscores, got: "partialBBQ". The application name is inferred from the path, if you'd like to explicitly name the application then use the "--app APP" option

→ Sessions

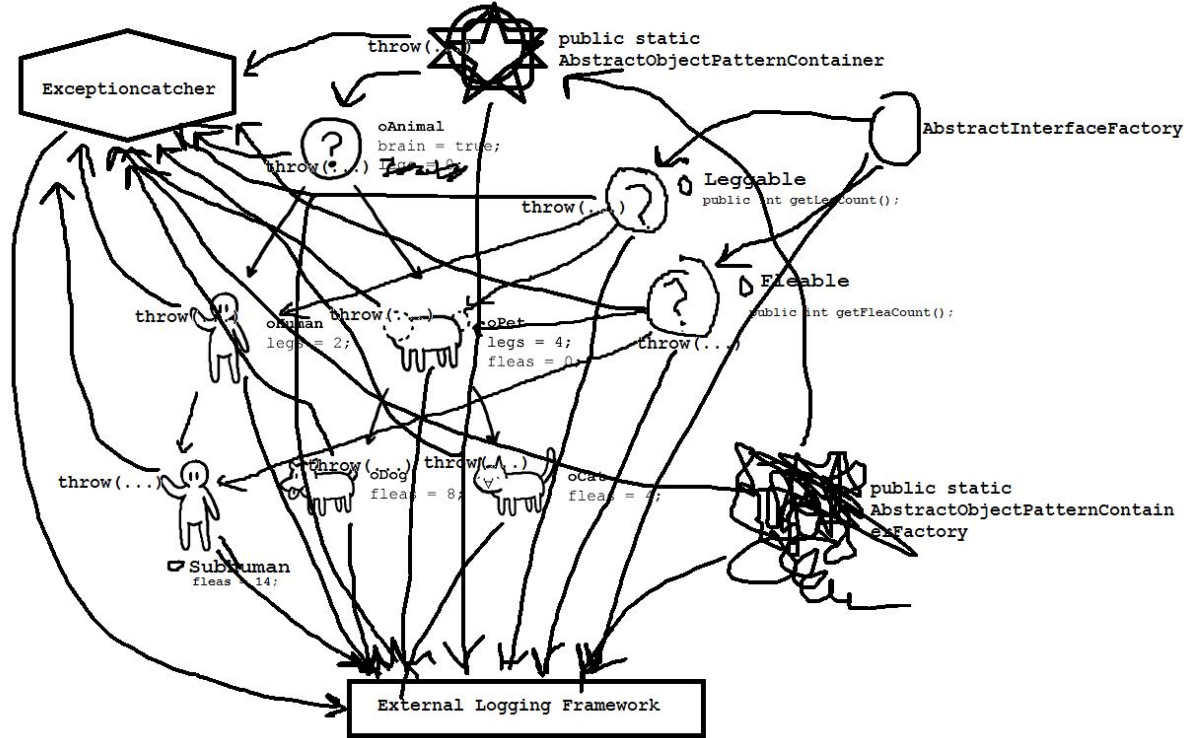
Why is Elixir awesome?

Functional

What OOP users claim



What actually happens



Why is Elixir awesome?

Standard Toolset

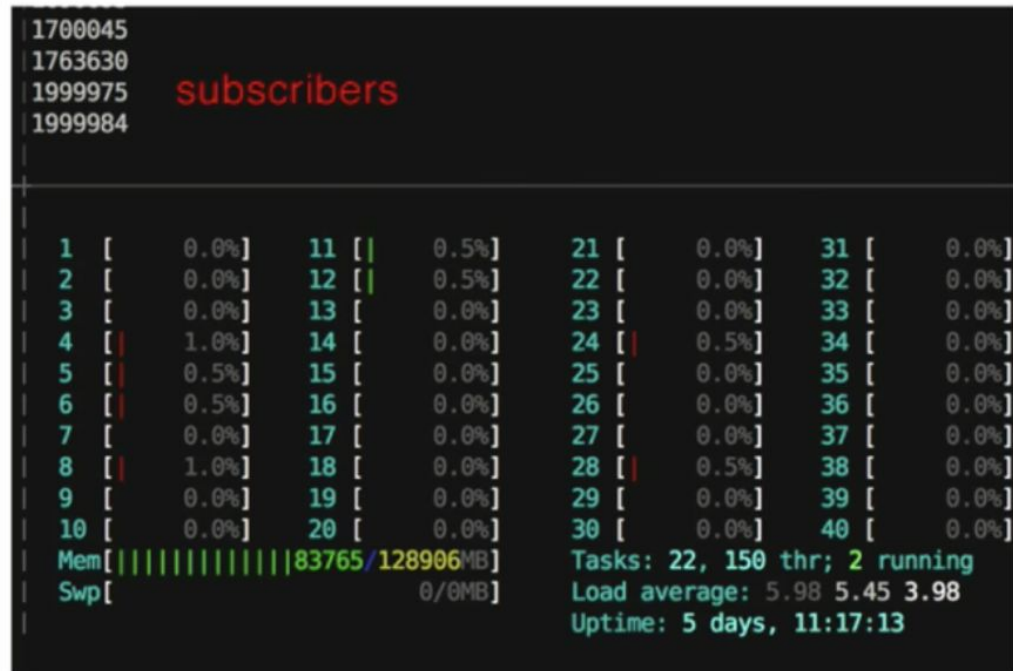
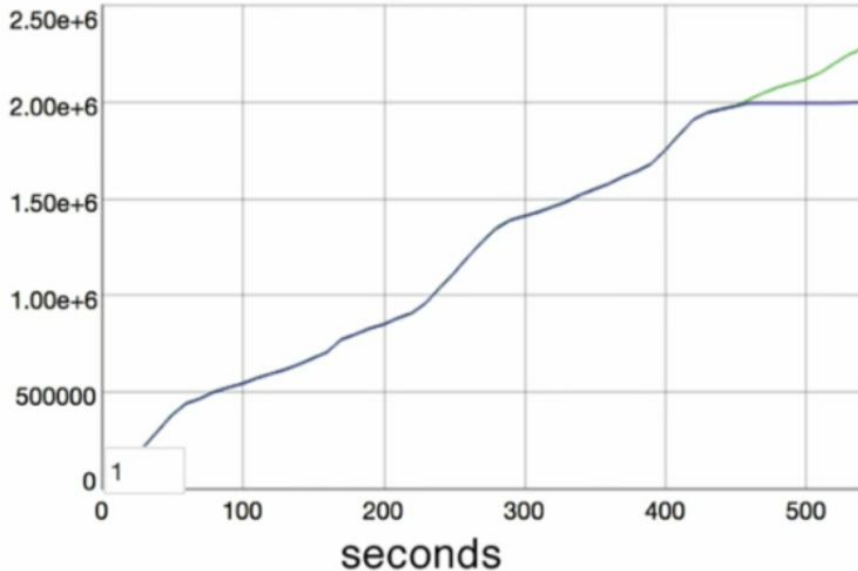
Why is Elixir awesome?

Builtin documentation

Why is Elixir awesome?

Scales like ...

Simultaneous Users



<https://www.youtube.com/watch?v=gom6nEvtl3U>

GOTO 2017 • Elixir: The only Sane Choice in an Insane World • Brian Cardarella

Why is Elixir awesome?

Scales like ...



About 10 engineers

PBs : Partial BBQ's

What runs Elixir ?

What runs Elixir ?

Erlang/OTP

Deep Dive : Concepts

Deep Dive : Concepts

Process

Process

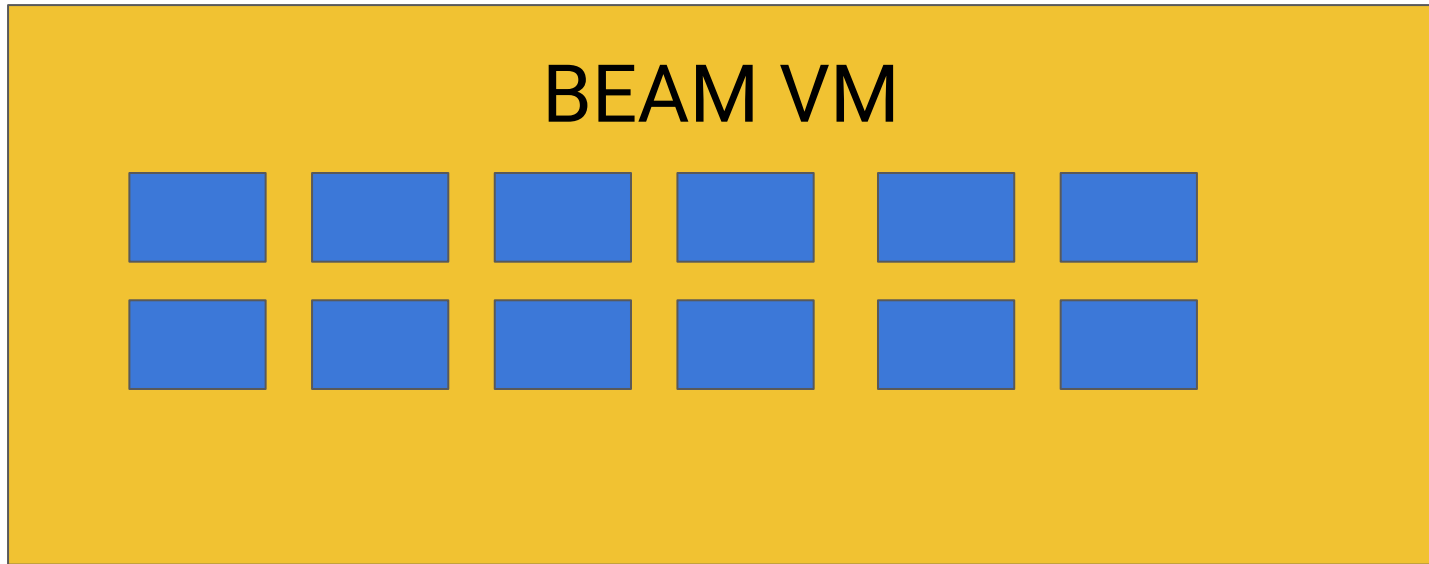
Process A

abc()
def()
ghi()
...

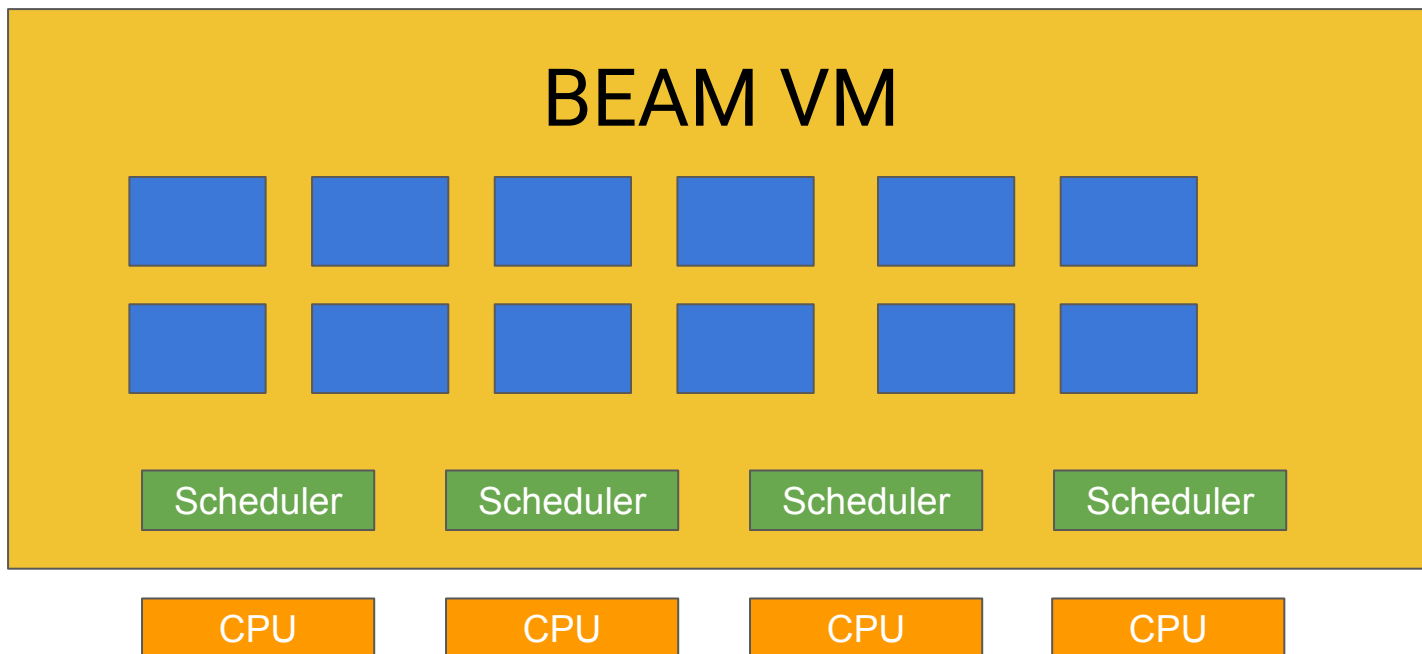
Process B

foo()
bar()
baz()
...

Process

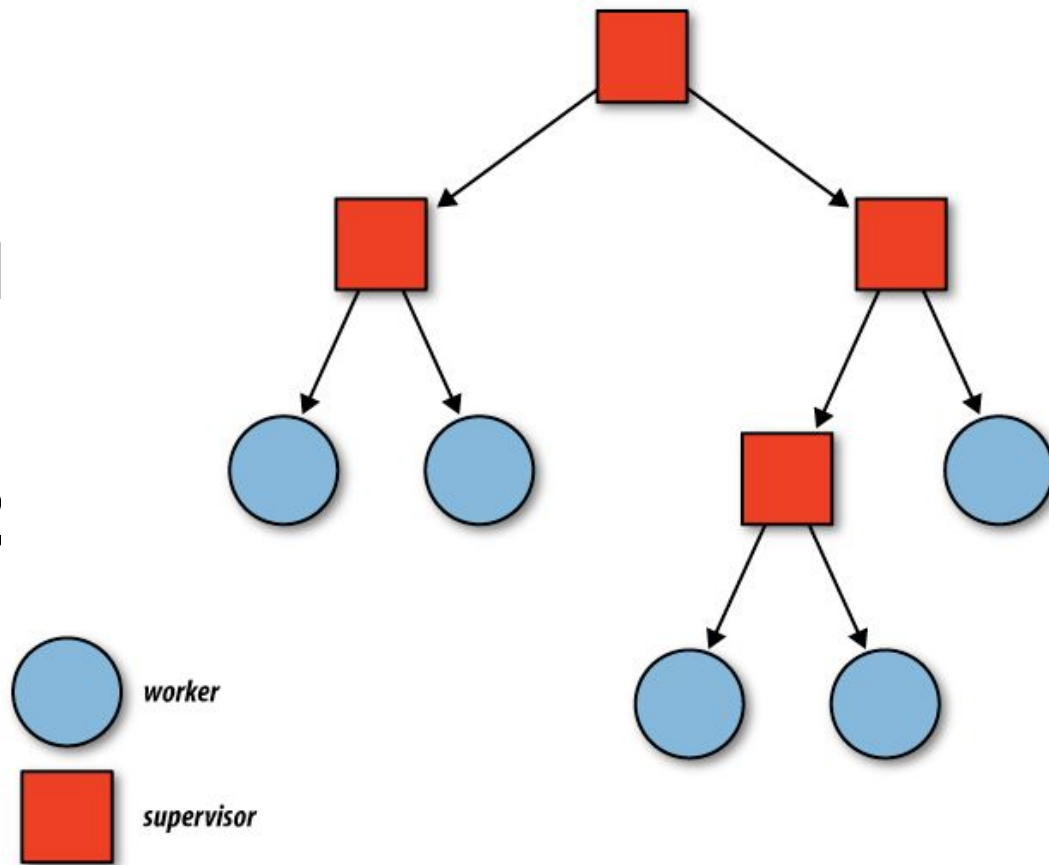


Process



Deep Dive : Concepts

Supervisor



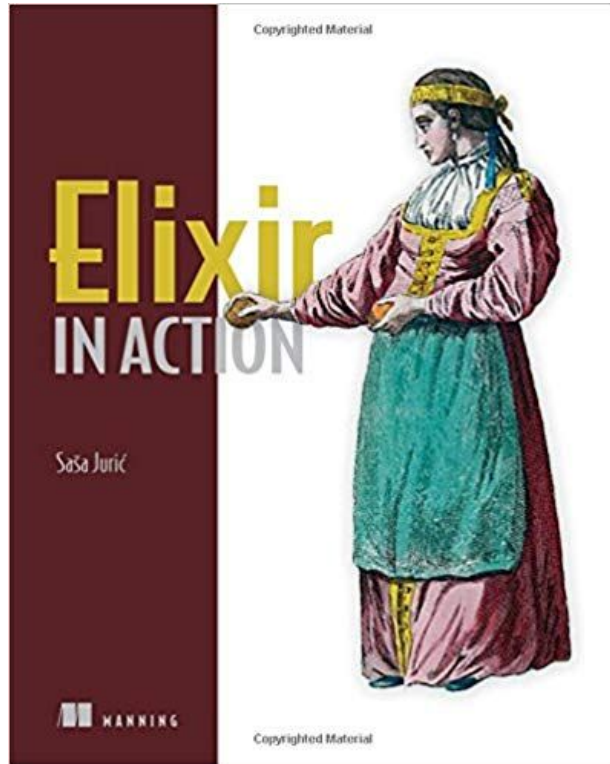
<https://virviil.github.io/2017/10/25/elixir-brackets-hell-in-supervision-tree/>

Technical requirement	Server A
HTTP server	Nginx and Phusion Passenger
Request processing	Ruby on Rails
Long-running requests	Go
Server-wide state	Redis
Persistable data	Redis and MongoDB
Background jobs	Cron, Bash scripts, and Ruby
Service crash recovery	Upstart

Technical requirement	Server A	Server B
HTTP server	Nginx and Phusion Passenger	Erlang
Request processing	Ruby on Rails	Erlang
Long-running requests	Go	Erlang
Server-wide state	Redis	Erlang
Persistable data	Redis and MongoDB	Erlang
Background jobs	Cron, Bash scripts, and Ruby	Erlang
Service crash recovery	Upstart	Erlang

Start simple

Go far



Elixir in Action

GoPlaces

GoPlaces